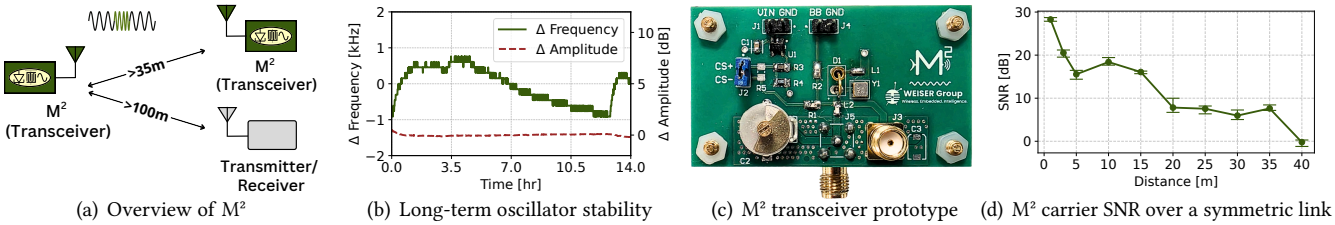
Demo: Enabling Low-power Data Streaming through Microwatt Microwave Transceivers

Dhairya Shah*[1], Pramuka Sooriya Patabandige*[1], Rajashekar Reddy Chinthalapani[1], Spanddhana Sara[1], Prabal Dutta[2], Ambuj Varshney[1]

[1]National University of Singapore [2]UC Berkeley

{dhairya, pramuka, rajashekar.c, spanddhanasara}@u.nus.edu, prabal@berkeley.edu, ambujv@nus.edu.sg



(a) Overview of M² (b) Long-term oscillator stability (c) M² transceiver prototype (d) M² carrier SNR over a symmetric link

Figure 1: (a) M² breaks the power–stability tradeoff in microwave oscillator design, enabling standalone microwatt transceivers. (b) It achieves 2.10 ppm **frequency stability at** 434.22 MHz **over** 14 hours**, with a maximum drift of only** 916 Hz **under** 105 µW**. (c) M² transceiver prototype used in this demonstration. (d) Two M² nodes form a symmetric link to enable tag-to-tag communication beyond** 35 m **in a complex indoor environment.**

Abstract

Stable carrier generation at microwatt power has long been considered incompatible with frequency stability. M² overcomes this limitation by coupling a tunnel diode with a high-Q SAW resonator, generating a microwave carrier at just 105 µW with 2.10 ppm stability over extended durations. Because the same circuit exhibits self-oscillating mixing and autodyning, a single M² board serves as both transmitter and receiver, enabling a symmetric link that requires no external emitter device. We demonstrate low-power data streaming across such a link, where a sensor modulates the M² transmitter directly with no amplification or digitization, and an M² receiver placed tens of meters away receives it. In addition, a live dashboard shows the stability of M² in real-time, while operating in a dynamically changing environment.

CCS Concepts

• **Hardware → Sensor devices and platforms**; **Wireless devices**; *Networking hardware*; • **Computer systems organization → Sensor networks**.

Keywords

Embedded systems, Low-power communication, Wireless transceivers, Tunnel diodes, SAW resonators

∗ Co-primary authors with equal contributions to this work.

This work is licensed under a Creative Commons Attribution 4.0 International License.

MobiCom '26, Austin, TX, USA

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2505-0/2026/10
https://doi.org/10.1145/3795866.3848446

ACM Reference Format:
Dhairya Shah, Pramuka Sooriya Patabandige, Rajashekar Reddy Chinthalapani, Spanddhana Sara, Prabal Dutta, and Ambuj Varshney. 2026. Demo: Enabling Low-power Data Streaming through Microwatt Microwave Transceivers. In *The 32nd Annual International Conference on Mobile Computing and Networking (MobiCom '26), October 26–30, 2026, Austin, TX, USA.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3795866.3848446

1 Introduction

Radio transceivers remain the most power-consuming component in embedded systems [5, 6]. Analog components such as oscillators, mixers, and amplifiers dominate their power consumption and have not benefited from Moore's Law to the same extent as their digital counterparts. Microwave oscillators are central to this challenge, since commodity wireless standards require high-frequency precision and stability, previously achievable only at milliwatt-scale power.

This constraint led to the delegation architecture, where power-hungry carrier generation is transferred to externally powered infrastructure [7, 17, 18]. In backscatter, an emitter device (ED) generates the carrier while the tag modulates it at very low power. However, the resulting three-device topology (ED, tag, receiver) has hindered widespread adoption beyond RFID. A true two-device architecture on a microwatt budget was considered impossible, since phase-locked loops draw milliwatts and microwatt tunnel diode oscillators (TDOs) lack the required stability at UHF [9, 10].

Streaming media compounds this cost at both ends. Battery-free streaming systems [8, 16] reach low-power operation only by delegating the carrier to an ED. Moreover, weak analog sensor signals are typically amplified and digitized before reaching the radio. These stages often dominate the sensor energy budget, motivating recent work on computation-free sensing [1]. M² removes both: it generates a stable carrier

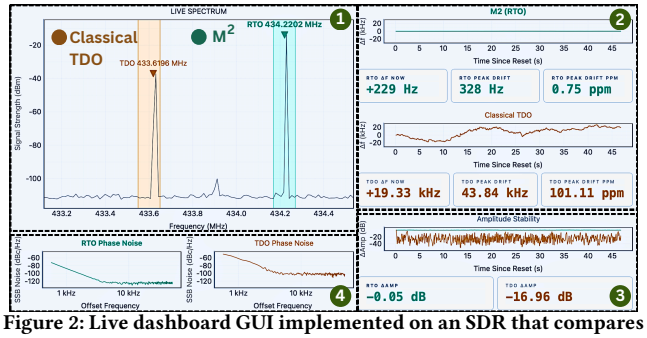


Figure 2: Live dashboard GUI implemented on an SDR that compares stability of M² against a classical TDO and reports drift figures in ppm. Sections – ❶: live spectrum showing both carriers, ❷: frequency stability, ❸: amplitude stability, and ❹: phase noise.

locally that is modulated directly by weak analog or digital baseband signals without dedicated amplifiers or mixers.

M² [12–14] couples a tunnel diode with a high-Q SAW resonator to generate a 433 MHz carrier at 105 µW with 2.10 ppm stability over 14 hours (Figure 1(b)). This stability holds under motion, voltage, and temperature variations, at least an order of magnitude better than related systems [2, 10]. Self-oscillating mixing (SoM) and autodyning properties let a single M² node transmit and receive over 100 m, and two nodes form a symmetric link beyond 35 m (Figures 1(a), 1(d)). We demonstrate live stability and a data streaming link.

2 Design and Evaluation

Stability. The quality factor (Q) of an oscillator's resonant circuit governs its frequency stability, phase noise, and consequent link quality. Classical TDOs rely on low-Q LC tanks whose environmentally coupled parasitics limit Q below 100. Thus, the carrier frequency drifts with temperature, humidity, and antenna loading. Although recent efforts [2, 9, 10] reduce drift through bias control and shielding, they still rely on inherently low-Q LC tanks. A SAW resonator instead confines energy in a mechanical standing wave on a piezoelectric substrate, setting resonant frequency lithographically by electrode pitch, yielding Q above 1000 and attenuating the frequency drift caused by a given perturbation.

Circuit. M² couples a Ge 1N3712 tunnel diode to a 433.92 MHz SAW resonator through a custom matching and biasing network. The SAW resonator fixes the oscillation frequency through acoustic resonance, while the diode, biased in its negative differential resistance region, sustains oscillation. The matching network transforms the diode's impedance to the SAW's motional impedance and cancels the total circuit reactance within its narrow inductive window; a variable capacitor aligns this cancellation point, without which fabrication tolerances prevent oscillation. The frontend (Figure 1(c)) including the SAW resonator consumes 105 µW.

Live stability dashboard. Carrier signals from M² and a TDO are tracked on a commodity ADALM-PLUTO SDR in real time under identical ambient conditions on a dashboard (Figure 2). It visualizes the spectrum and stability, showing M² is orders of magnitude more stable than TDO.

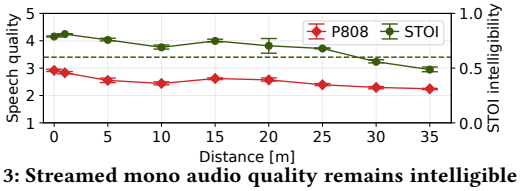


Figure 3: Streamed mono audio quality remains intelligible up to 25 m over a symmetric M²-to-M² link in an indoor LoS environment.

Transceiver. The matching network preserves the diode's nonlinear I-V characteristic, enabling the same circuit to function as a transceiver. In transmission, it enables SoM: a baseband signal (f_b) is upconverted to $f_0 \pm f_b$ without a separate mixer. The baseband signal can come directly from a sensor or from an MCU stepped down to below 60 mV. In reception, autodyning [7] mixes an incoming RF signal (f_1) near f_0 with its carrier and is downconverted to $|f_0 - f_1|$. We tested the transmitter up to 100 kbps with 2-FSK at tens of meters, limited by the commodity receiver. The nonlinearity supports even higher frequencies and data rates at both ends.

Data streaming. A passive MEMS microphone modulates the carrier directly with zero additional power at the transmitter node. An identical receiver node generates its carrier at a nominal offset within the SAW bandwidth set by board variations [14]. The incoming signal is downconverted to the offset frequency, which is fairly stable for a node pair. The demodulation pipeline (an RC filter, an amplifier, and a digitizer) adds about 200 µW [7]. For audio, the amplifier drives a speaker directly. We evaluate the link in a complex indoor LoS environment by playing a standard test audio [4] into the microphone and recording the downconverted audio at increasing distances, three runs each, scored offline for speech quality with DNSMOS P.808 [3, 11] and intelligibility with STOI [15]. Speech quality falls to 2.3 at 35 m, while intelligibility remains above 0.6 threshold up to 25 m (Figure 3). While Figure 1(d) reports detectable carrier SNR up to 35 m, intelligible audio beyond 25 m demands a higher SNR.

3 Demonstration

We demonstrate M²'s live oscillator stability at the booth alongside a TDO. Visitors perturb both by hand gestures or by moving nearby, and compare the resulting changes in frequency, amplitude, and phase noise. We also show media transmission and reception on a symmetric M² to M² audio streaming link. A visitor speaks into the microphone, and the recovered speech is interpreted at the receiver to perform a task such as querying an LLM and displaying its response. Together, these demonstrate oscillator stability, transceiver operation, and data streaming in a dynamic environment.

Acknowledgements. This work is funded by an ODPRT Startup Grant (A-8000277-00-00) and unrestricted gifts from Google through its Research Scholar Program (A-8002307-00-00-00) and TPU Builders Program (A-8004887-00-00), hosted at the National University of Singapore. This work is partly supported by an Okawa Foundation Research Grant and DENSO International America, Inc., hosted at UC Berkeley.

References

[1] Rajashekar Reddy Chinthalapani, Pramuka Sooriya Patabandige, Dhairya Shah, Savitha Viswanadh Kandala, Spanddhana Sara, and Ambuj Varshney. 2026. WiP: ZeCoST: Zero-Compute Streaming Sensors using Tunnel Diodes. In *Proceedings of the 32nd Annual International Conference on Mobile Computing and Networking* (Austin, TX, USA) *(MobiCom '26)*. ACM, New York, NY, USA, 1–4. doi:10.1145/3795866.3844791 [To appear].

[2] Rajashekar Reddy Chinthalapani, Dhairya Jigar Shah, Nobel Ang, and Ambuj Varshney. 2025. AudioCast: Enabling Ubiquitous Connectivity for Embedded Systems through Audio-broadcasting Low-power Tags. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 2, Article 27 (June 2025), 32 pages. doi:10.1145/3729471

[3] ITU-T. 2018. *Recommendation P.808: Subjective evaluation of speech quality with a crowdsourcing approach*. Recommendation P.808. International Telecommunication Union, Geneva. https://www.itu.int/rec/T-REC-P.808/en

[4] ITU-T. 2025. *Recommendation P.501: Test signals for use in telephonometry*. Recommendation P.501. International Telecommunication Union. https://www.itu.int/rec/T-REC-P.501-202504-I/en

[5] Bryce Kellogg, Vamsi Talla, Shyamnath Gollakota, and Joshua R. Smith. 2016. Passive Wi-Fi: Bringing Low Power to Wi-Fi Transmissions. In *13th USENIX Symposium on Networked Systems Design and Implementation (NSDI 16)*. USENIX Association, Santa Clara, CA, 151–164. https://www.usenix.org/conference/nsdi16/technical-sessions/presentation/kellogg

[6] Vincent Liu, Aaron Parks, Vamsi Talla, Shyamnath Gollakota, David Wetherall, and Joshua R. Smith. 2013. Ambient backscatter: wireless communication out of thin air. In *Proceedings of the ACM SIGCOMM 2013 Conference* (Hong Kong, China) *(SIGCOMM '13)*. Association for Computing Machinery, New York, NY, USA, 39–50. doi:10.1145/2486001.2486015

[7] Sooriya Patabandige Pramuka Medaranga, Rajashekar Reddy Chinthalapani, Wenqing Yan, Prabal Dutta, and Ambuj Varshney. 2025. Unraveling the Missing Link in Low-power Communication: An Autodyning Receiver Architecture that Achieves a Long Range. In *The 23rd Annual International Conference on Mobile Systems, Applications and Services (MobiSys '25)*. Association for Computing Machinery, Anaheim, CA, USA, 1–14. doi:10.1145/3711875.3729164

[8] Saman Naderiparizi, Mehrdad Hessar, Vamsi Talla, Shyamnath Gollakota, and Joshua R Smith. 2018. Towards Battery-Free HD Video Streaming. In *15th USENIX Symposium on Networked Systems Design and Implementation (NSDI 18)*. USENIX Association, Renton, WA.

[9] Madhushanka Padmal. 2025. *Wireless Communication Systems for Energy-Constrained Environments*. Ph. D. Dissertation. Uppsala University, Sweden. https://uu.diva-portal.org/smash/record.jsf?pid=diva2%3A1993808

[10] Madhushanka Padmal, Dilushi Piumwardane, and Thiemo Voigt. 2026. SharpPeak: Unlocking the True Potential of Tunnel Diodes for Low-Power Long-Range Communication. In *Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems (SenSys '26)*. Association for Computing Machinery, New York, NY, USA, 29–43. doi:10.1145/3774906.3800468

[11] Chandan K. A. Reddy, Vishak Gopal, and Ross Cutler. 2021. DNSMOS: A Non-Intrusive Perceptual Objective Speech Quality Metric to Evaluate Noise Suppressors. In *ICASSP 2021 – IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 6493–6497. doi:10.1109/ICASSP39728.2021.9414878

[12] Dhairya Shah, Pramuka Sooriya Patabandige, Rajashekar Reddy Chinthalapani, Spanddhana Sara, Prabal Dutta, and Ambuj Varshney. 2026. Demo Abstract: Microwatt Microwave (M2) Oscillator: Enabling 105 µW, Stable and Standalone Transceivers. In *Proceedings of the ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED '26)*. Association for Computing Machinery, New York, NY, USA, Article 63, 2 pages. doi:10.1145/3816440.3822367

[13] Dhairya Shah, Pramuka Sooriya Patabandige, Rajashekar Reddy Chinthalapani, Spanddhana Sara, Prabal Dutta, and Ambuj Varshney. 2026. Poster: Going Beyond the Delegation Architecture of Low-power Wireless Communication. In *Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services Companion* (University of Cambridge, Cambridge, United Kingdom) *(MobiSys Companion '26)*. Association for Computing Machinery, New York, NY, USA, 66–67. doi:10.1145/3812835.3814882

[14] Pramuka Sooriya Patabandige, Dhairya Shah, Rajashekar Reddy Chinthalapani, Spanddhana Sara, Prabal Dutta, and Ambuj Varshney. 2026. Microwatt Microwave (M^2) Oscillator: Going Beyond the Delegation Architecture of Low-power Wireless Communication. In *Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services* (University of Cambridge, Cambridge, United Kingdom) *(MobiSys '26)*. Association for Computing Machinery, New York, NY, USA, 941–959. doi:10.1145/3745756.3809248

[15] Cees H. Taal, Richard C. Hendriks, Richard Heusdens, and Jesper Jensen. 2011. An Algorithm for Intelligibility Prediction of Time-Frequency Weighted Noisy Speech. *IEEE Transactions on Audio, Speech, and Language Processing* 19, 7 (2011), 2125–2136. doi:10.1109/TASL.2011.2114881

[16] Vamsi Talla, Bryce Kellogg, Shyamnath Gollakota, and Joshua R. Smith. 2017. Battery-Free Cellphone. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 2, Article 25 (June 2017), 20 pages. doi:10.1145/3090090

[17] Ambuj Varshney, Oliver Harms, Carlos Pérez-Penichet, Christian Rohner, Frederik Hermans, and Thiemo Voigt. 2017. LoRea: A Backscatter Architecture that Achieves a Long Communication Range. In *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems* (Delft, Netherlands) *(SenSys '17)*. Association for Computing Machinery, New York, NY, USA, Article 18, 14 pages. doi:10.1145/3131672.3131691

[18] Ambuj Varshney, Wenqing Yan, and Prabal Dutta. 2022. Judo: addressing the energy asymmetry of wireless embedded systems through tunnel diode based wireless transmitters. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services* (Portland, Oregon) *(MobiSys '22)*. Association for Computing Machinery, New York, NY, USA, 273–286. doi:10.1145/3498361.3538923